

Tri fusion (Merge-sort) - Récursion illustrée

Étapes de la récursion

Algorithmes et Pensée Computationnelle

UNIL — École des Sciences Criminelles

Code

```
1 def sort(arr):  
2     if len(arr) <= 1:  
3         return arr  
4  
5     mid = len(arr) // 2  
6     left = sort(arr[:mid])  
7     right = sort(arr[mid:])  
8  
9     return merge(left, right)
```

Listing 1: Merge sort — Python

La fonction `merge` prend deux listes triées comme paramètres et renvoie une liste triée qui est la fusion de ces deux listes. La fonction `merge` a donc une complexité de $O(n)$.

Étape 1

```
sort([8,4,7,10,6,1,2,5])  
return:
```

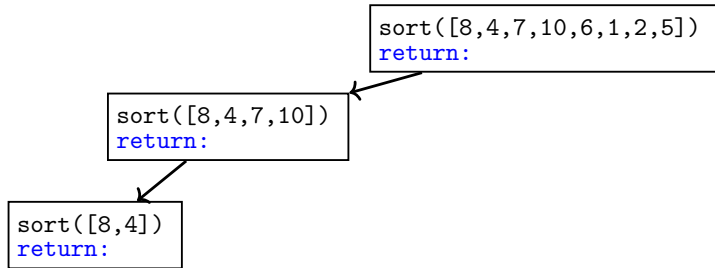
Étape 2

```
sort([8,4,7,10,6,1,2,5])  
return:
```

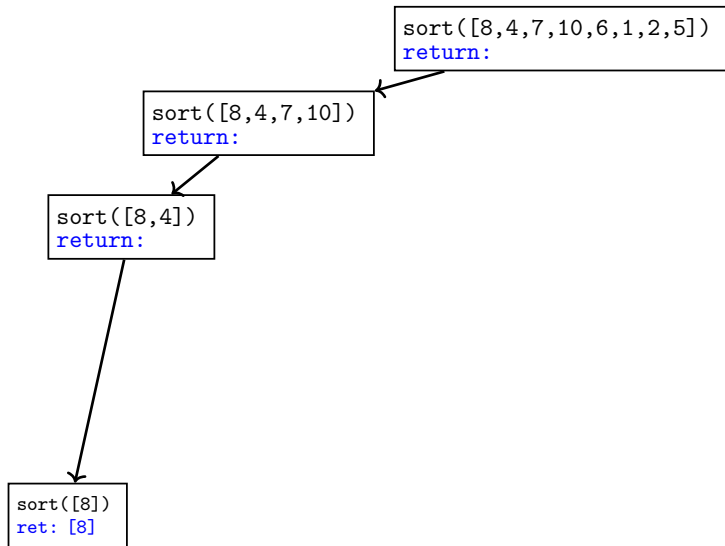
```
sort([8,4,7,10])  
return:
```



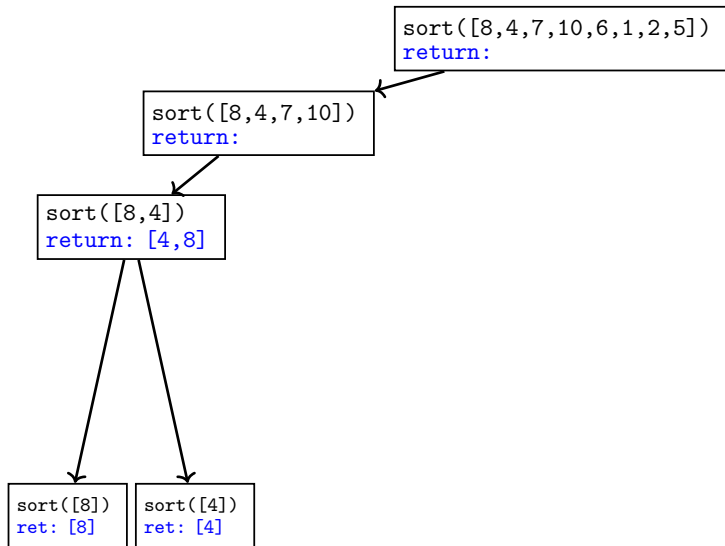
Étape 3



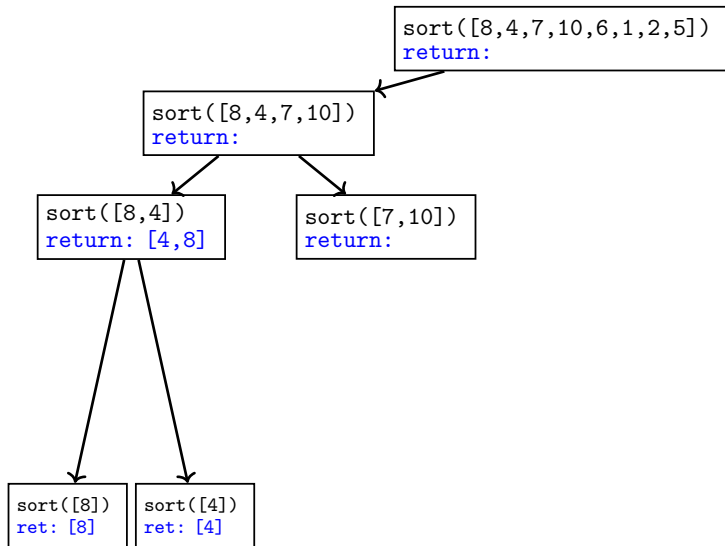
Étape 4



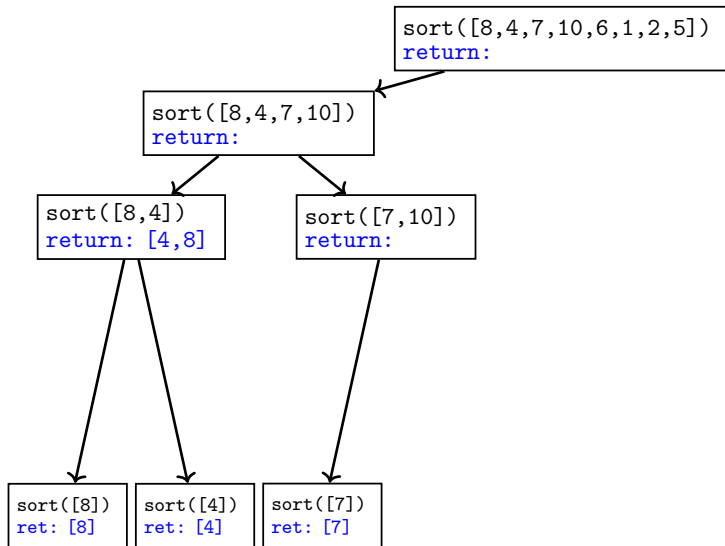
Étape 5



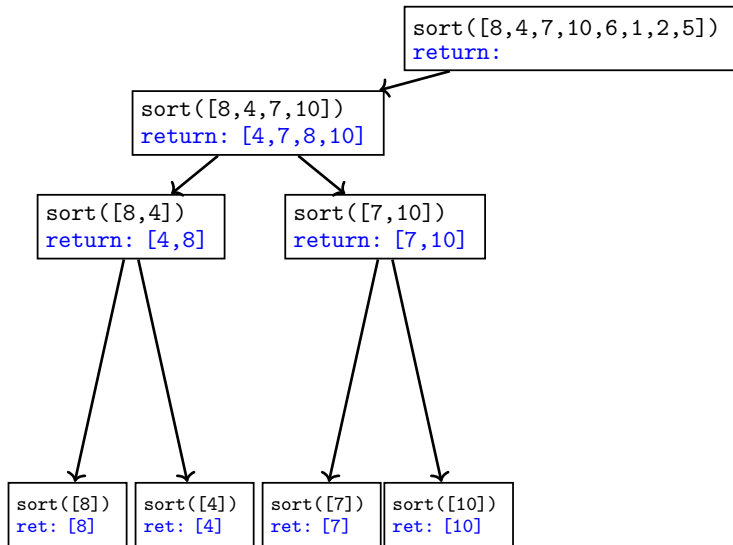
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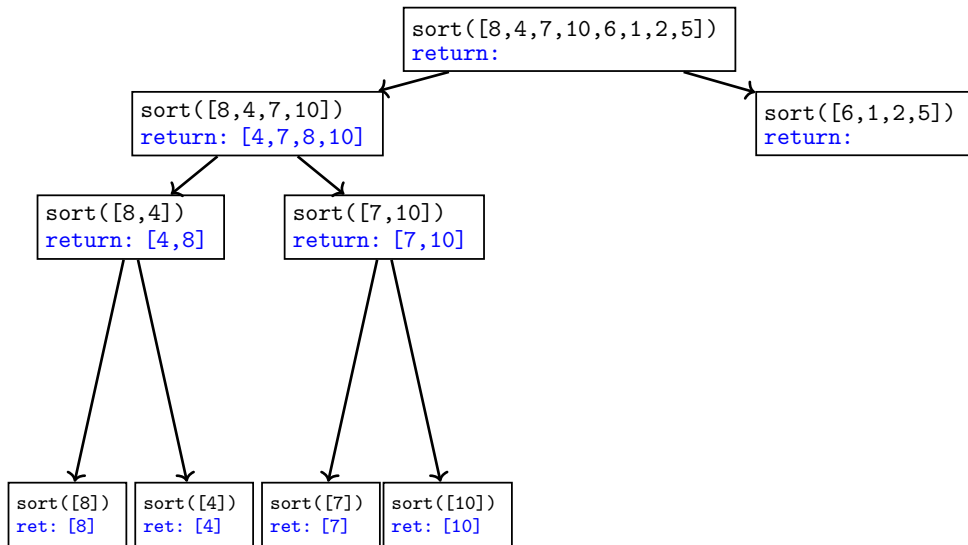
Étape 7



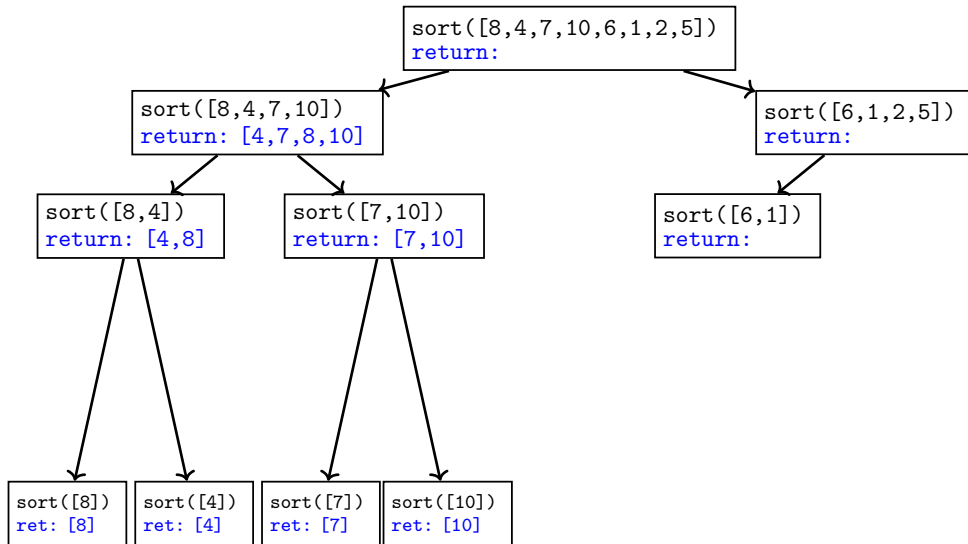
Étape 8



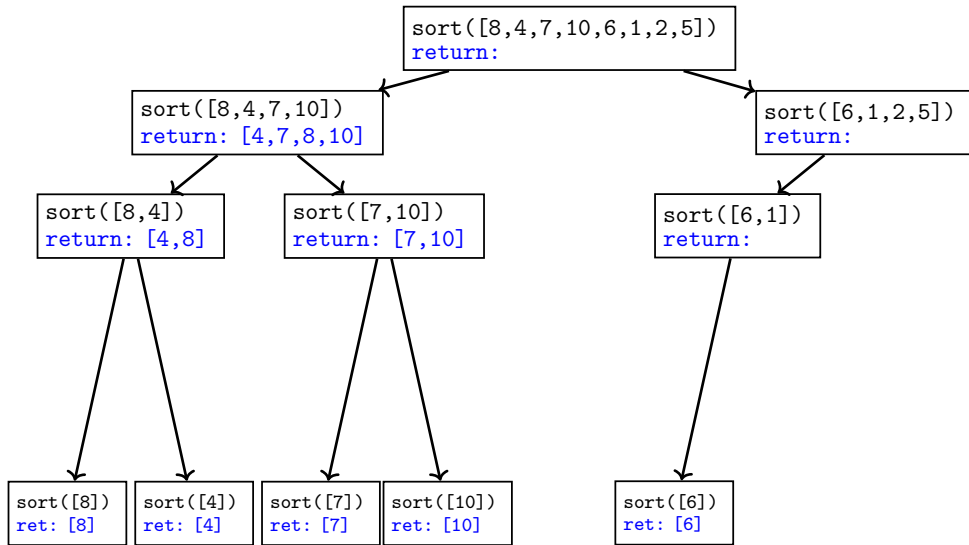
Étape 9



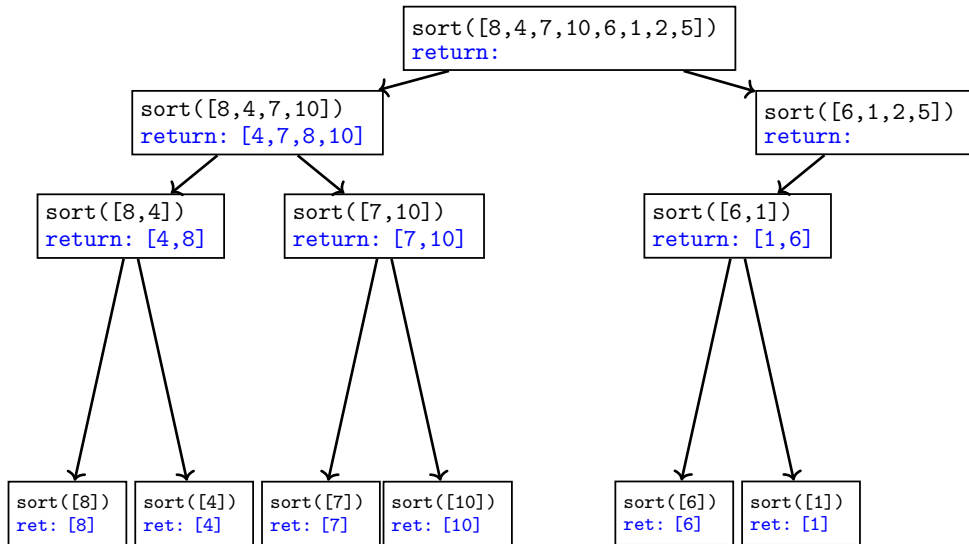
Étape 10



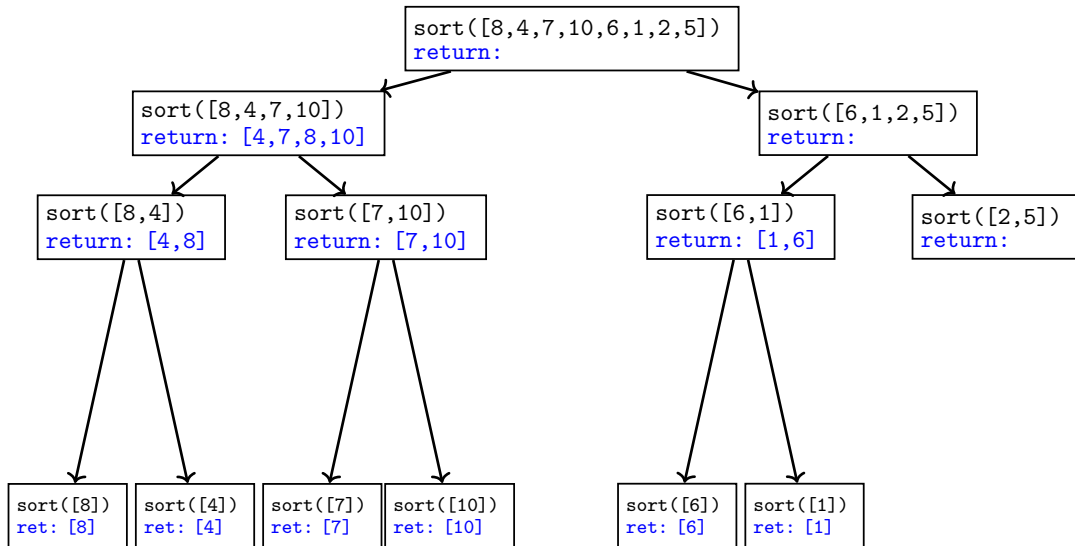
Étape 11



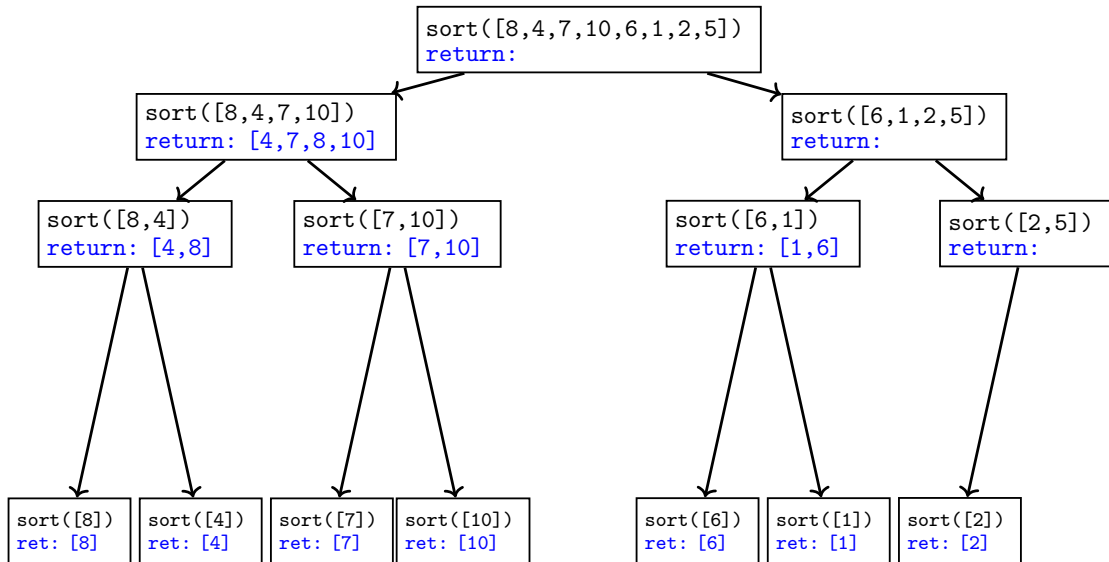
Étape 12



Étape 13



Étape 14



Étape 15

